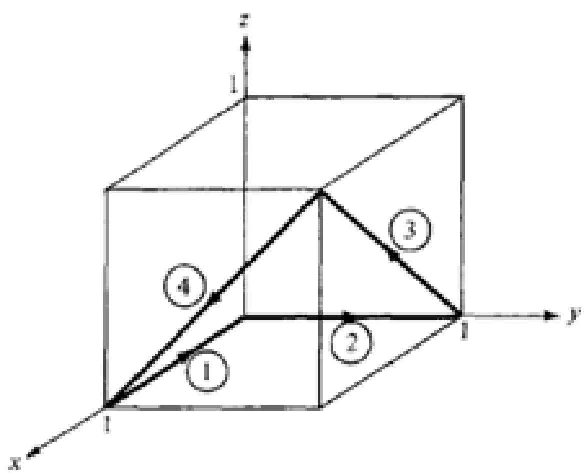
Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 3-Solutions

1. The circulation of $\boldsymbol{F}$ around path $\boldsymbol{L}$ is given by:

$$\oint_L \boldsymbol{F} \cdot d\boldsymbol{l} = \left(\int_{\mathbf{1}} + \int_{\mathbf{2}} + \int_{\mathbf{3}} + \int_{\mathbf{4}}\right) \boldsymbol{F} \cdot d\boldsymbol{l}$$

where the path is broken into segments numbered 1 to 4 as shown in the figure below.



For segment 1: y = 0 = z, $\boldsymbol{F} = x^2 \boldsymbol{a_x}$

and

$$d\boldsymbol{l} = dx\ \boldsymbol{a_x}$$

Notice that $d\boldsymbol{l}$ is always taken as along $+\boldsymbol{a_x}$ so that the direction on segment 1 is taken care of by the limits of integration. Thus,

$$\int_{\mathbf{1}} \boldsymbol{F}.d\boldsymbol{l} = \int_{\mathbf{1}}^{\mathbf{0}} x^2 dx = \left.\frac{x^3}{3}\right|_1^0 = -\frac{1}{3}$$

For segment 2, $x = 0 = z$:

$$\boldsymbol{F} = -y^2 \boldsymbol{a_z}$$

and

$$d\boldsymbol{l} = dy\ \boldsymbol{a_y}$$

$$\boldsymbol{F}.\boldsymbol{dl} = 0$$

hence:

$$\int_{\mathbf{2}} \boldsymbol{F}.d\boldsymbol{l} = 0$$

For segment 3, $y = 1$,

$$\boldsymbol{F} = x^2 \boldsymbol{a_x} - xz\ \boldsymbol{a_y} - \boldsymbol{a_z}$$

and

$$d\boldsymbol{l} = dx\ \boldsymbol{a_x} + dz\ \boldsymbol{a_z}$$

$$\int_{\mathbf{3}} \boldsymbol{F}.d\boldsymbol{l} = \int (x^2 dx - dz)$$

But on 3, $z = x$,

$$dx = dz$$

hence:

$$\int_{\mathbf{3}} \boldsymbol{F}.d\boldsymbol{l} = \int_0^1 (x^2 - 1)dx = \left.\frac{x^3}{3} - x\right|_0^1 = -\frac{2}{3}$$

For segment 4, $x = 1$,

$$\boldsymbol{F} = \boldsymbol{a_x} - z\,\boldsymbol{a_y} - y^2\,\boldsymbol{a_z}$$

and

$$d\boldsymbol{l} = dy\,\boldsymbol{a_y} + dz\,\boldsymbol{a_z}$$

hence:

$$\int_{\mathbf{4}} \boldsymbol{F}.d\boldsymbol{l} = \int (-zdy - y^2dz)$$

But on 4, $z = y$,

$$dz = dy$$

so:

$$\int_{\mathbf{3}} \boldsymbol{F}.d\boldsymbol{l} = \int_1^0 (-y - y^2)dy = -\frac{y^2}{2} - \frac{y^3}{3}\bigg|_1^0 = \frac{5}{6}$$

By putting all these together, we obtain:

$$\oint_L \boldsymbol{F}\,.\,d\boldsymbol{l} = -\frac{1}{3} + 0 - \frac{2}{3} + \frac{5}{6} = -\frac{1}{6}$$

2. a.

$$\overline{\nabla}\boldsymbol{U} = \frac{\partial U}{\partial x}\boldsymbol{a_x} + \frac{\partial U}{\partial y}\boldsymbol{a_y} + \frac{\partial U}{\partial z}\boldsymbol{a_z}$$

$$\overline{\nabla}\boldsymbol{U} = \mathbf{4}z^2\boldsymbol{a_x} + 3z\boldsymbol{a_y} + (8xz + 3y)\boldsymbol{a_z}$$

b.

$$\overline{\nabla}\boldsymbol{W} = \frac{\partial W}{\partial \rho}\boldsymbol{a}_\rho + \frac{\partial W}{\partial \phi}\boldsymbol{a}_\phi + \frac{\partial W}{\partial z}\boldsymbol{a_z}$$

$$\overline{\nabla}\boldsymbol{W} = 2(z^2 + 1)\cos\phi\,\boldsymbol{a}_\rho - 2(z^2 + 1)\sin\phi\,\boldsymbol{a}_\phi + 4\rho z\cos\phi\,\boldsymbol{a_z}$$

c.

$$\overline{\nabla}\boldsymbol{H} = \frac{\partial H}{\partial r}\boldsymbol{a}_r + \frac{\partial H}{\partial \theta}\boldsymbol{a}_\theta + \frac{\partial H}{\partial \phi}\boldsymbol{a}_\phi$$

$$\overline{\nabla}\boldsymbol{H} = 2r\cos\theta\cos\phi\,\boldsymbol{a}_r - r\sin\theta\cos\phi\,\boldsymbol{a}_\theta - r\cot\theta\sin\phi\,\boldsymbol{a}_\phi$$

3. a.

$$\nabla.\boldsymbol{A} = ye^{\mathrm{xy}} + x\cos xy - 2x\cos zx\sin zx$$

$$\nabla\times\boldsymbol{A} = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{\mathrm{xy}} & \sin xy & cos^2\,xz \end{vmatrix}$$

$$\nabla\times\boldsymbol{A} = (\,0-0)\,\boldsymbol{a}_x + (\boldsymbol{0} + 2\boldsymbol{z}\cos zx\sin zx)\,\boldsymbol{a}_y + (y\cos xy - xe^{\mathrm{xy}})\boldsymbol{a}_z$$

$$\nabla\times\boldsymbol{A} = z\sin 2xz\;\;\boldsymbol{a}_y + (y\cos xy - xe^{\mathrm{xy}})\;\boldsymbol{a}_z$$

b.

$$\nabla.\boldsymbol{B} = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho^2 z^2\cos\phi) + 0 + sin^2\,\phi$$

$$\nabla.\boldsymbol{B} = 2z^2\cos\phi + sin^2\,\phi$$

$$\nabla\times\boldsymbol{B} = \left(\frac{1}{\rho}\frac{\partial B_{\boldsymbol{z}}}{\partial\phi} - 0\right)\boldsymbol{a}_\rho + \left(\frac{\partial B_\rho}{\partial z} - \frac{\partial B_z}{\partial\rho}\right)\boldsymbol{a}_\phi + \frac{1}{\rho}\left(0 - \frac{\partial B_\rho}{\partial\phi}\right)\boldsymbol{a}_z$$

$$\nabla\times\boldsymbol{B} = \frac{z\sin 2\phi}{\rho}\boldsymbol{a}_\rho + 2\rho z\cos\phi\,\boldsymbol{a}_\phi + z^2\sin\phi\,\boldsymbol{a}_z$$

c.

$$\nabla.\boldsymbol{C} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^3\cos\theta) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\left(-\frac{1}{r}sin^2\,\phi\right) + 0$$

$$\nabla.\boldsymbol{C} = 3\cos\theta - \frac{2\cos\theta}{r^2}$$

$$\nabla\times\boldsymbol{C} = \frac{1}{r\sin\theta}\left(\frac{\partial}{\partial\theta}(2r^2\,sin^2\,\theta) - 0\right)\boldsymbol{a}_r$$
$$+\frac{1}{r}\left(0 - \frac{\partial}{\partial r}(2r^3\sin\theta)\right)\boldsymbol{a}_\theta\;\frac{1}{r}\left(\frac{\partial}{\partial r}(-\sin\theta) + r\sin\theta\right)\boldsymbol{a}_\phi$$

$$\nabla\times\boldsymbol{C} = 4r\cos\theta\,\boldsymbol{a}_r - 6r\sin\theta\,\boldsymbol{a}_\theta + \sin\theta\,\boldsymbol{a}_\phi$$